



ETRO
ELECTRONICS &
INFORMATICS

TASK 2: Process Explorer and Swap Memory on a macOS System

OPERATING SYSTEMS AND SECURITY (OSSEC)

- **Student Name:** Zinar Mutlu
- **Student ID:** 0629829
- **Email:** zinar.mutlu@vub.be
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Introduction

The objective of this task is to examine active processes and memory management on a macOS system. This includes an overview of tools for process monitoring, network connection tracking, and swap memory management. Instead of Windows tools like Task Manager and Sysinternals Utilities, I will use macOS-specific tools such as **Activity Monitor**, `top`, and `htop` to explore active processes and manage swap memory efficiently. The goal is to understand how macOS handles system resources and how users can analyze and optimize system performance.

1. Exploring Active Processes in macOS

Activity Monitor is a built-in macOS utility that provides a graphical interface for monitoring system performance and managing processes. For more detailed system monitoring and resource management, macOS provides command-line tools like `top` and `htop`.

- `top`: A built-in macOS tool for real-time process monitoring.
- `htop`: An enhanced version of `top`, offering an interactive and user-friendly interface.

1.1 Activity Monitor (GUI Tool)

- Access via **Applications** → **Utilities** → **Activity Monitor** or Spotlight (`Cmd + Space` , then search for "Activity Monitor").

Activity Monitor

Activity Monitor

All Processes

ⓧ

ℹ

⋮

CPU

Memory

Energy

Disk

Network

🔍

Search

Process Name	% CPU	CPU Time	Threads	Idle Wake-Ups	Kind	% GPU	GPU Time	PID
Vivaldi Helper (Renderer)	29,0	2:06,91	18	4	Apple	0,0	0,00	11547
WindowServer	16,4	46:37,58	22	99	Apple	1,0	9:13,15	186
kernel_task	8,7	20:00,54	559	1059	Apple	0,0	0,00	0
Vivaldi	8,6	8:35,18	43	3	Apple	0,0	0,00	11525
Boom 3D	8,1	16:59,85	10	0	Intel	0,0	0,00	866
coreaudiod	7,0	15:17,46	14	99	Apple	0,0	0,00	198
Code Helper (Plugin)	5,8	11:47,16	20	72	Apple	0,0	0,00	10000
Vivaldi Helper (Renderer)	4,1	11:48,72	20	20	Apple	0,0	0,00	13064
StocksWidget	3,8	2:37,37	10	1	Apple	0,0	0,00	849
Acrobat Pro DC	3,2	1:36,87	15	99	Intel	0,0	0,01	22226
CleanMyMac X HealthMonitor	3,1	1:59,19	12	6	Apple	0,0	0,00	819
iStat Menus Status	2,0	4:23,56	7	4	Apple	0,0	0,00	837
Activity Monitor	1,9	3,81	6	5	Apple	0,0	0,00	32982
Accessibility Services	1,9	2:32,10	5	11	Apple	0,0	0,00	655
Stocks	1,8	1:10,49	8	2	Apple	0,0	0,00	915
CleanMyMac X Menu	1,5	48,67	12	1	Apple	0,0	0,00	828
distnoted	1,2	53,21	4	0	Apple	0,0	0,00	457
Vivaldi Helper (GPU)	1,2	5:35,32	18	64	Apple	0,0	5:05,49	11531

System: %3,33

User: %6,41

Idle: %90,26

CPU LOAD



Threads: 3.819

Processes: 727

Activity Monitor (All Processes)

Monitors

Dock Icon

Options

Show All Windows

Hide

Quit

Show Activity Monitor

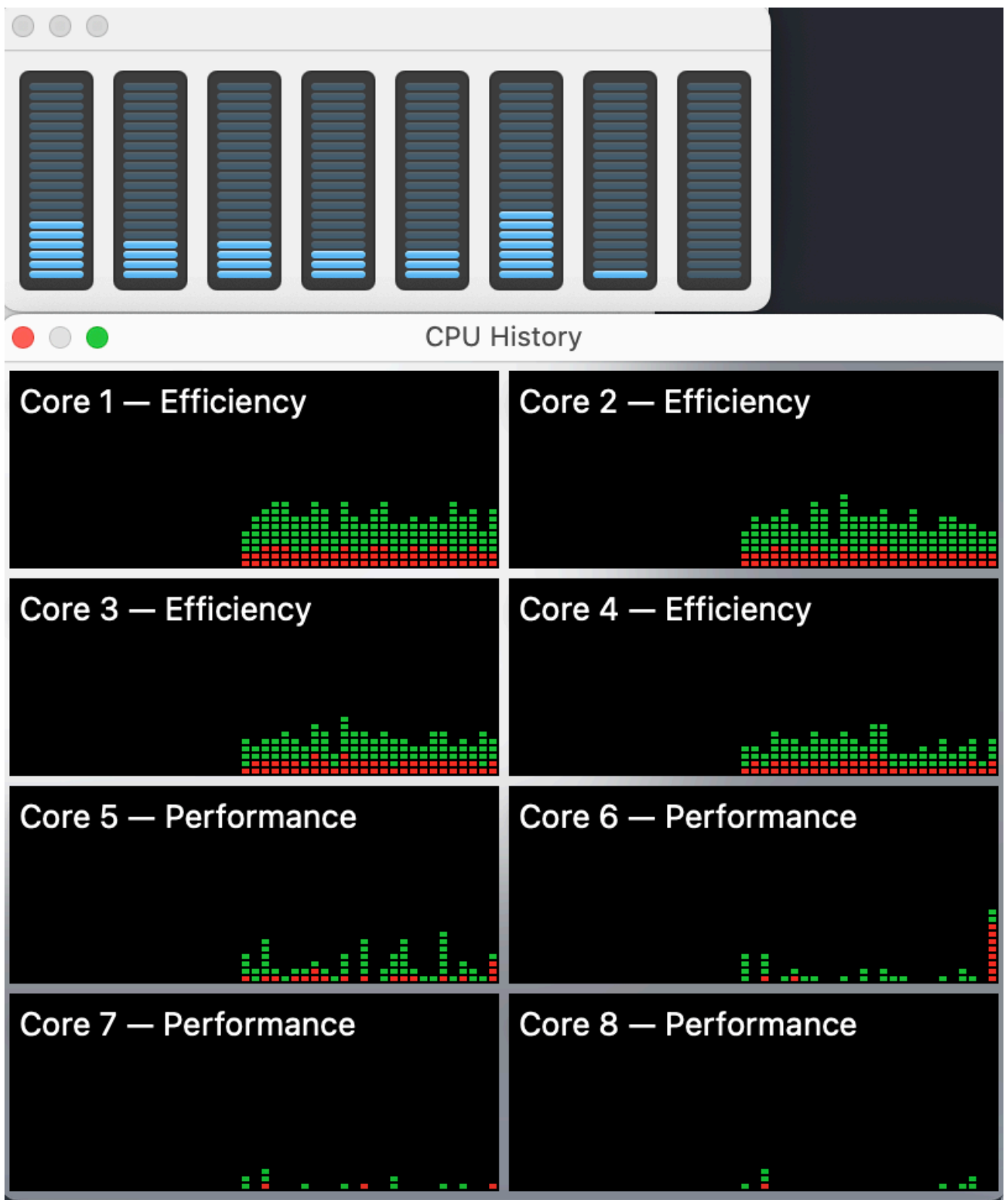
Show CPU Usage

Show CPU History

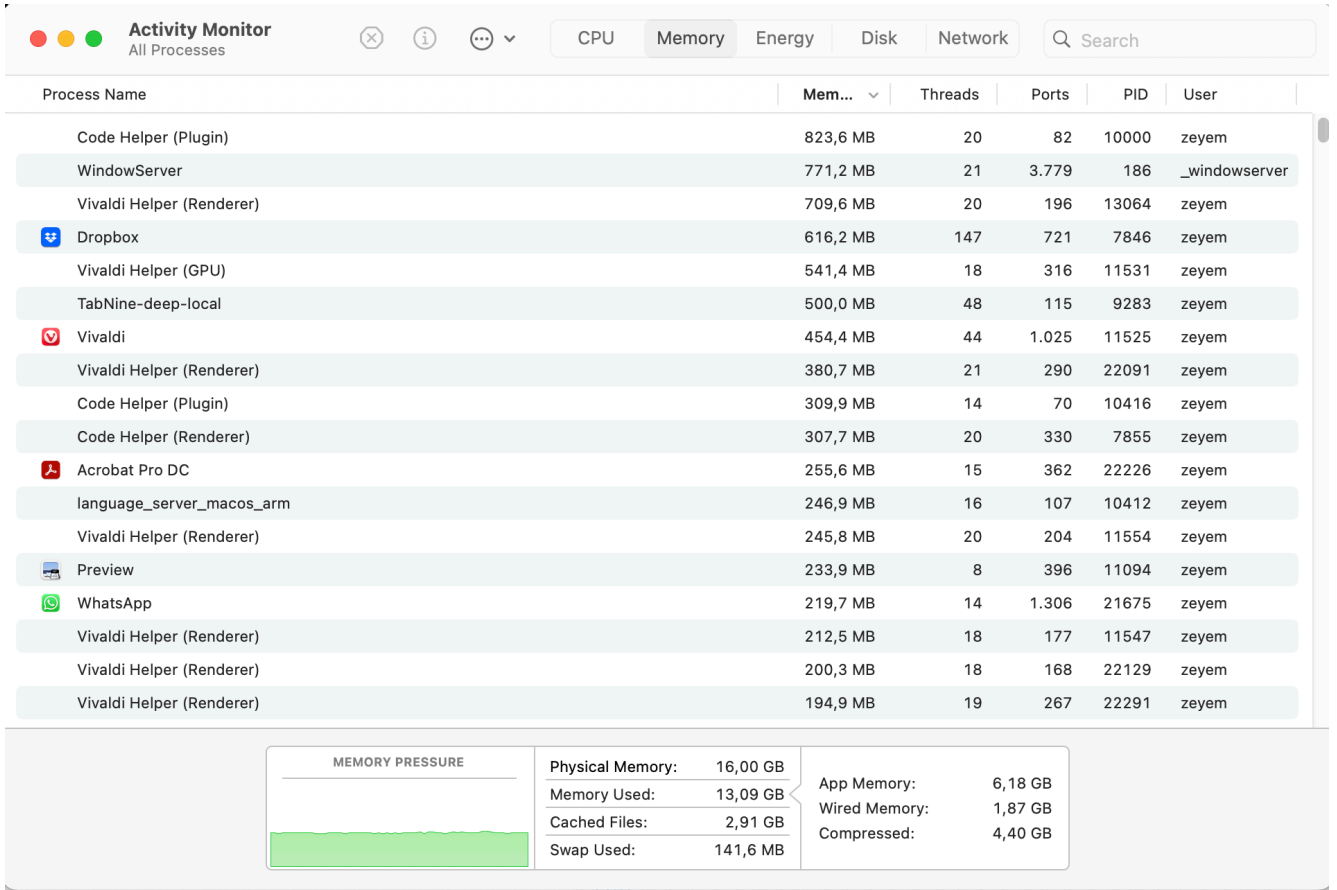
Clear CPU History

- Displays CPU, Memory, Energy, Disk, and Network usage for each running process.

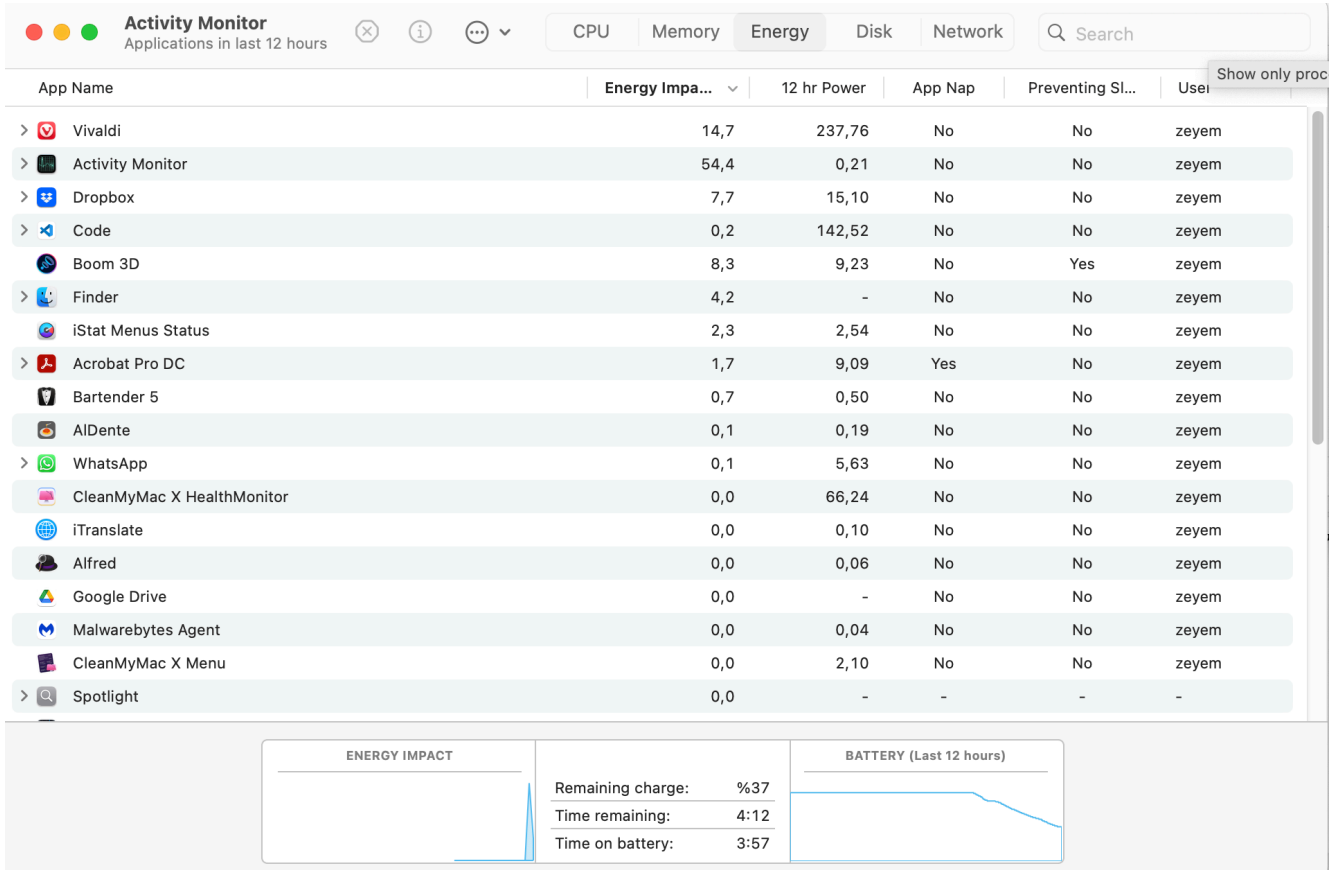
CPU Usage



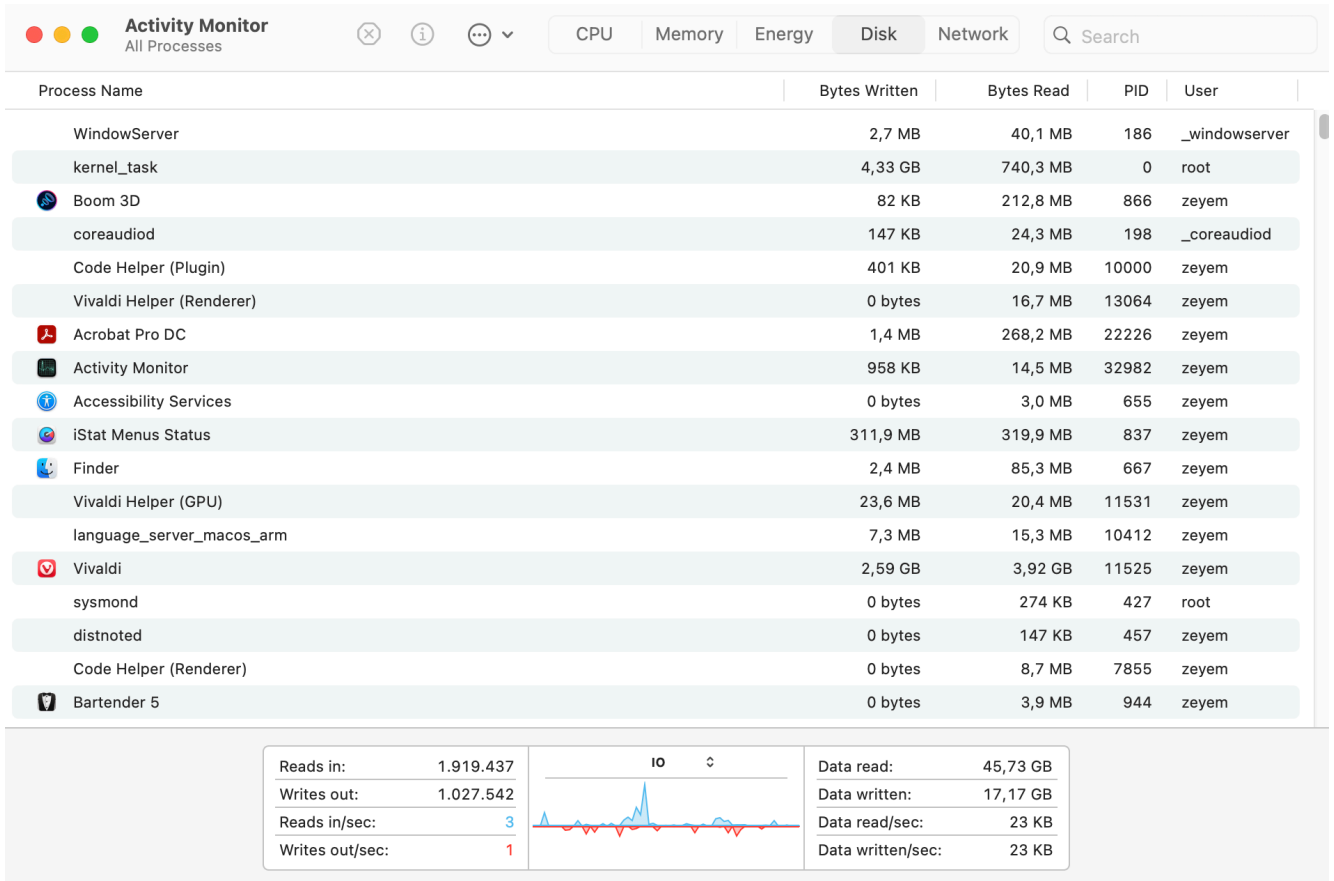
Memory



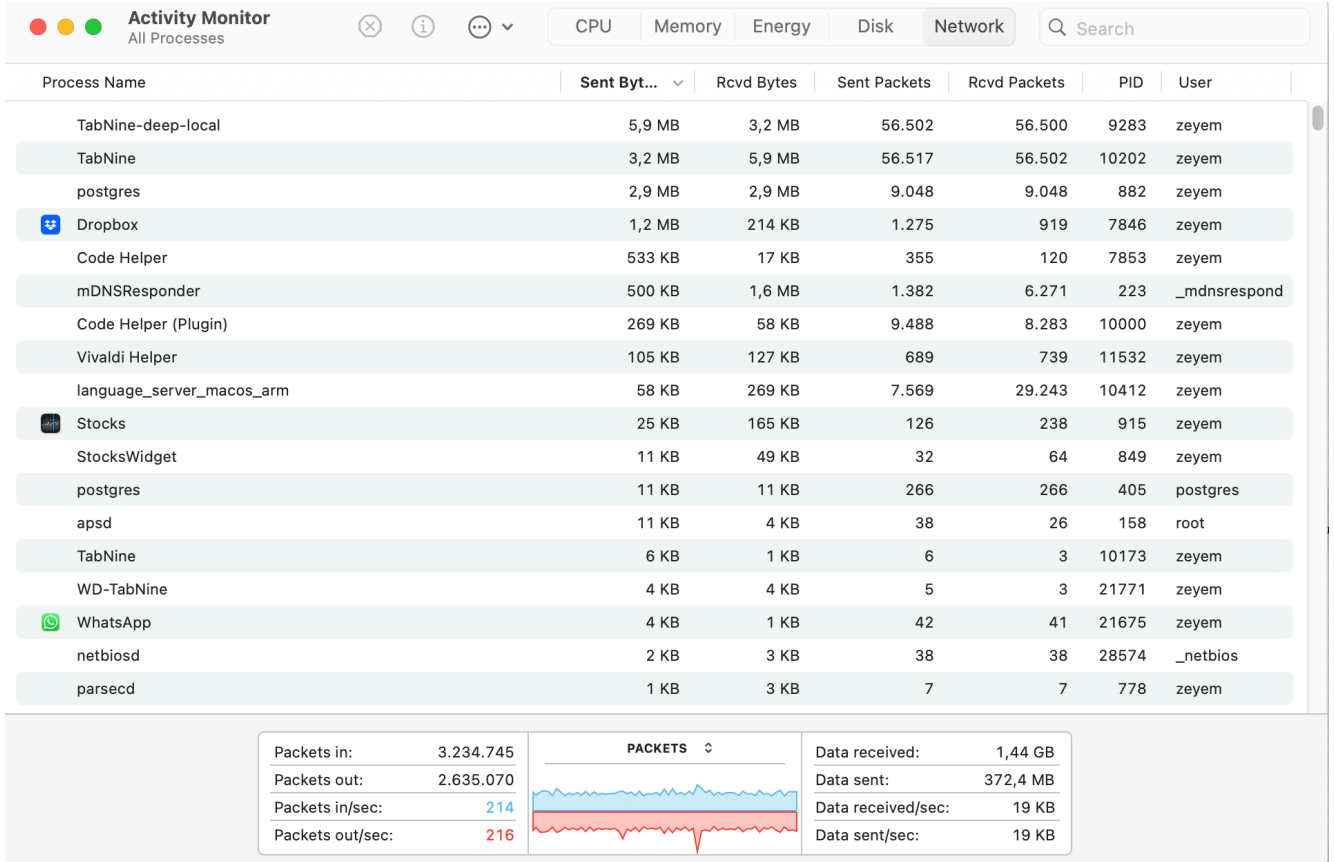
Energy



Disk



Netowrk



- Allows users to **force quit** applications or processes consuming excessive resources.

Activity Monitor
All Processes

CPUMemoryEnergyDiskNetwork

Search

Process Name	% CPU	CPU Time	Threads	Idle Wake-Ups	Kind	% GPU	GPU Time	PID
WindowServer	21,6	48:51,80	23	117	Apple	4,9	9:28,20	186
kernel_task	11,8	21:09,97	559	1300	Apple	0,0	0,00	0
Boom 3D	7,9	17:52,64	9	0	Intel	0,0	0,00	866
coreaudiod	6,9	16:05,08	12	100	Apple	0,0	0,00	198
Code Helper (Plugin)	5,6				Apple	0,0	0,00	10000
Acrobat Pro DC	3,3				Intel	0,0	0,01	22226
iStat Menus Status	2,4				Apple	0,0	0,00	837
Activity Monitor	2,2				Apple	0,0	0,00	32982
Accessibility Services	1,5				Apple	0,0	0,00	655
language_server_macos_arm	1,4				Apple	0,0	0,00	10412
screencapture	1,3				Apple	0,0	0,00	33841
iStatMenusDaemon	1,3				Apple	0,0	0,00	319
Bartender 5	1,2				Apple	0,1	4,69	944
Vivaldi	1,1				Apple	0,0	0,00	11525
sysmond	1,1				Apple	0,0	0,00	427
distnoted	1,0				Apple	0,0	0,00	457
Code Helper (Renderer)	0,9	9:22,43	20	1	Apple	0,0	0,00	7855
monad	0,8	1:25,33	34	31	Intel	0,0	0,00	817

Are you sure you want to quit this process?
Do you really want to quit "Boom 3D"?

Quit

Force Quit

Cancel

System: 3,87%

User: 8,22%

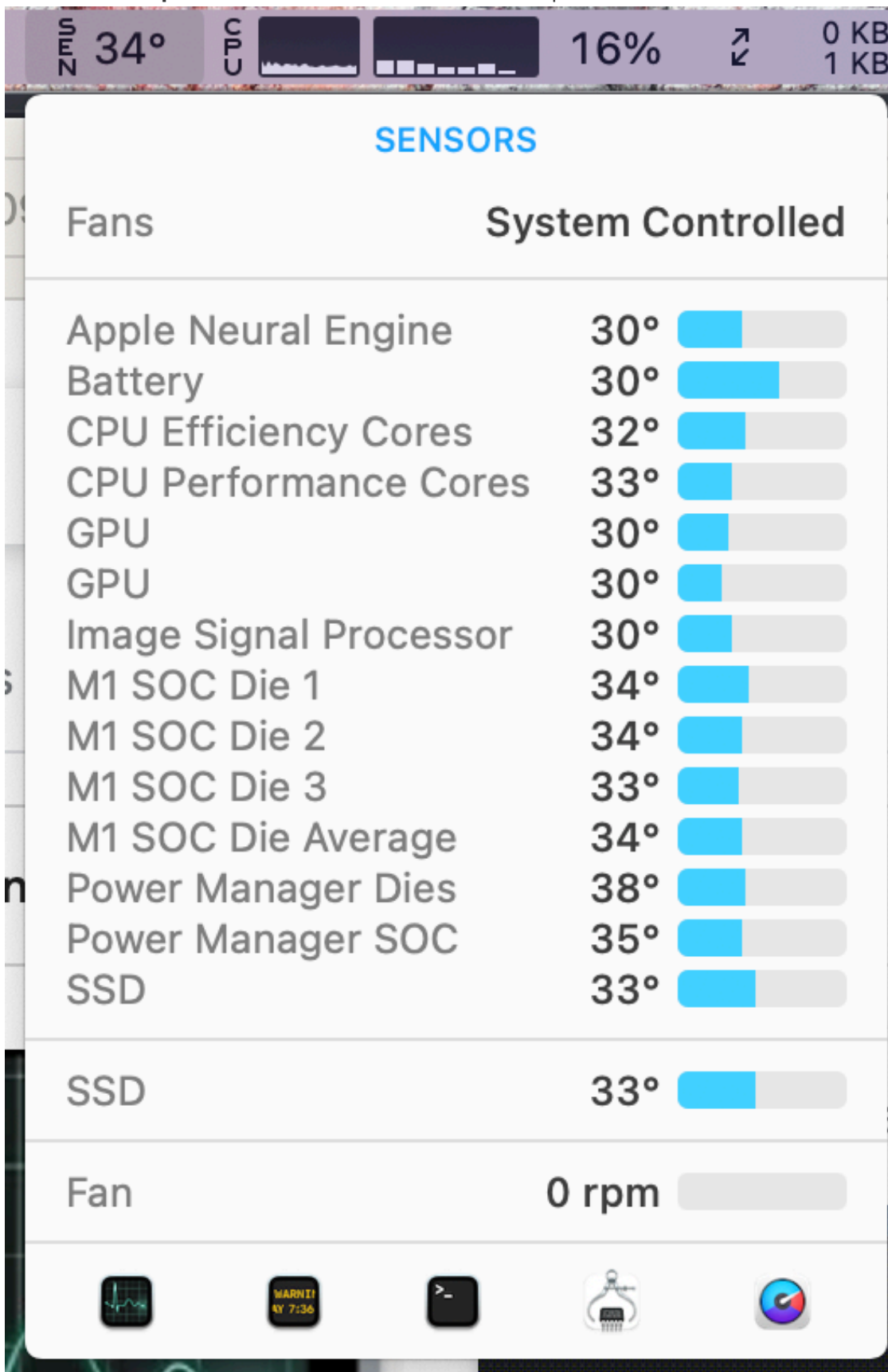
Idle: 87,91%

CPU LOAD

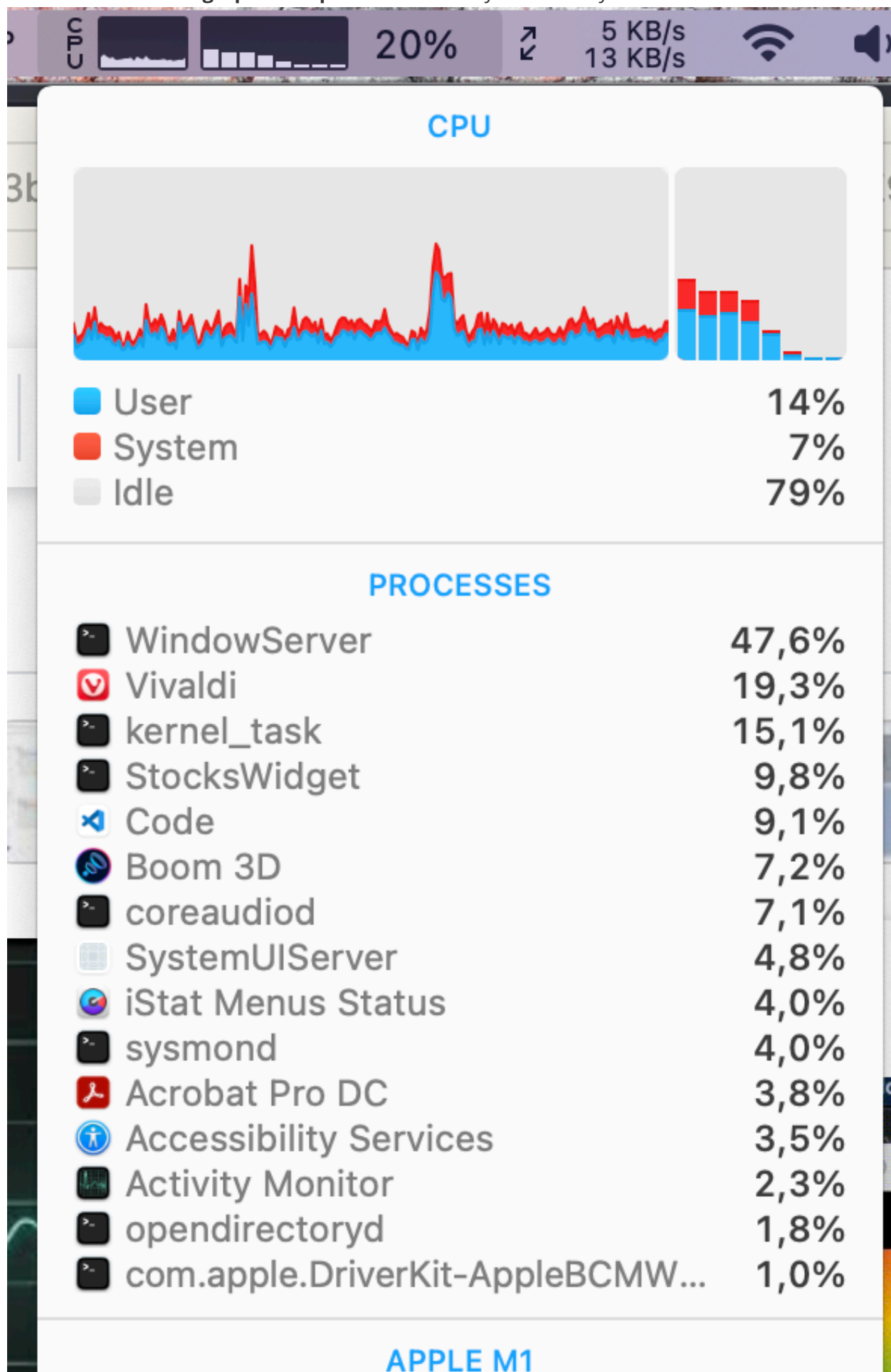
Threads: 3.824

Processes: 729

- Users can **sort processes** based on resource consumption.



- Provides a **real-time graphical representation** of system activity.



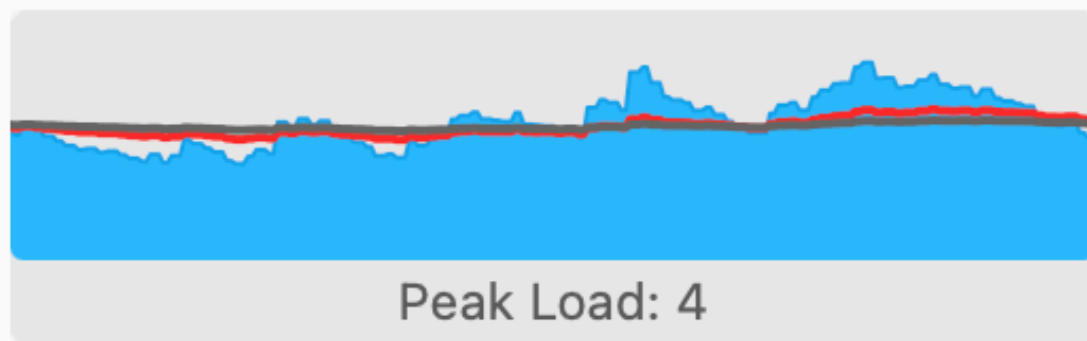
Memory
Processor



Frames Per Second

46,0

LOAD AVERAGE



■ 1,96

■ 2,24

■ 2,18

UPTIME

4 hours, 17 minutes



CPU History

Core 1 — Efficiency



Core 2 — Efficiency



Core 3 — Efficiency

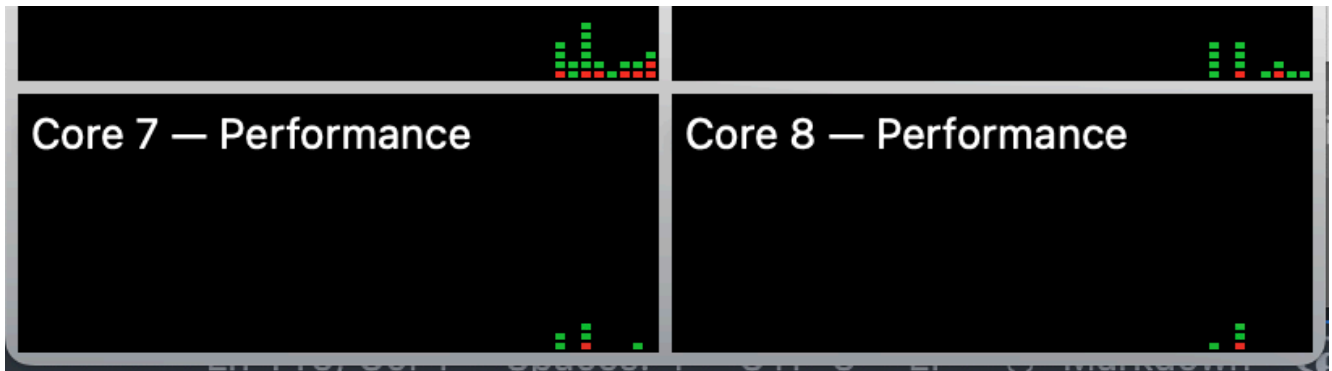


Core 4 — Efficiency



Core 5 — Performance

Core 6 — Performance



1.2 Terminal-Based Process Monitoring: `top` Command

Run in Terminal: `top`

```
Dropbox — top — 105x26
13:00:27
Processes: 734 total, 2 running, 732 sleeping, 3903 threads
Load Avg: 3.87, 2.63, 2.34 CPU usage: 7.64% user, 4.79% sys, 87.55% idle
SharedLibs: 831M resident, 217M data, 278M linkedit.
MemRegions: 0 total, 0B resident, 4720K private, 1833M shared.
PhysMem: 15G used (1860M wired, 5080M compressor), 116M unused.
VM: 320T vsize, 9207M framework vsize, 5361(0) swapins, 23400(0) swapouts.
Networks: packets: 3591104/1432M in, 2986478/414M out. Disks: 2052707/45G read, 1133236/17G written.

PID    COMMAND      %CPU    TIME    #TH    #WQ    #PORT  MEM    PURG    CMPRS    PGRP    PPID    STATE
186*   WindowServer 20.2    52:39.03 24     6      4037-  844M+  34M-    336M    186     1      sleeping
0*     kernel_task  12.8    22:55.93 559/8  0       0      3104K  0B      0B      0       0      running
866*   Boom 3D      9.0     19:07.37 9       1      374    69M-   0B      62M     866     1      sleeping
38080* top          8.1     00:02.23 2/1     0       37     10M    0B      0B      38080   37645  running
198*   coreaudioid  7.8     17:13.41 12      4      958    27M    0B      15M     198     1      sleeping
10000* Code Helper  6.1     13:33.49 20      1      82     828M+  0B      457M-   7844    7844    sleeping
7855*  Code Helper  4.0     10:58.07 20      1      332    318M+  0B      58M     7844    7844    sleeping
22226* AdobeAcrobat 3.5     02:31.87 15      4      362    256M   0B      183M    22226   1      sleeping
156*   opendirector 3.2     00:41.05 7       6      2162-  12M-   64K     6672K-  156     1      sleeping
38089* screencaptur 2.3     00:00.28 2       1      66     8690K  752K    0B      666     666     sleeping
655*   AXVisualSupp 2.3     03:03.05 5       1      224    15M    0B      7264K   655     1      sleeping
457*   distnoted    1.4     01:10.15 4       3      527    3969K  0B      1184K   457     1      sleeping
817*   mongod       1.4     01:31.21 34      1      82     41M    0B      28M-    817     1      sleeping
10412* language_ser 1.3     03:39.42 16      0      107    242M   0B      124M-   7844    10303   sleeping
196*   airportd     1.3     00:48.70 8       6      355    10M-   0B      3360K   196     1      sleeping
337*   com.apple.Dr 0.7     01:49.45 8       6      1460   30M    0B      4864K   337     1      sleeping
35169* Activity Mon 0.6     00:11.67 8       5      508    48M    96K     16M     35169   1      sleeping
```

- Displays real-time CPU and memory usage of running processes.
- Lists a list of active processes sorted by CPU usage.
- Shows memory consumption, process ID (PID), and execution time.
- Updates in real-time, providing dynamic process tracking.
-

zoyem - top - 229x48

Processes: 741 total, 2 running, 739 sleeping, 3895 threads

Load Avg: 2.36, 2.43, 2.32 CPU usage: 9.38% user, 6.38% sys, 84.24% idle SharedLibs: 833M resident, 218M data, 294M linkedit. MemRegions: 0 total, 0B resident, 0B private, 2081M shared.

PhysMem: 156 used (1871M wired, 5247M compressed), 190M unused. VM: 325T vmize, 928M framework vmize, 536k(0) swaptins, 2340k(0) swapouts. Networks: packets: 3640918/1447M in, 5831828/419M out.

Disks: 2871253/450 read, 1144042/170 written.

PID	COMMAND	%CPU	TIME	#TH	#WO	#PORT	MEM	PURG	CMPRS	PGRP	PPID	STATE	BOOSTS	NCPU	ME	NCPU_OTHERS	UID	FAULTS	COW	MSGSENT	MSGRECV	SYSSBD	SYVSMA	CSW	PAGEIN	IDLE	POWE
186+	WindowServer	12.2	53:23.94	22	6	4133+	867M-	45M+	339M	186	1	sleeping	*0(1)	0.00076	0.05600	88	5821857+	47358	20976875+	23249401+	48614596+	74432508+	17925339+	2976	582441+	12.7	
0+	kernel_task	12.0	23:18.07	559/9	0	0	3136K	0B	0B	0	0	running	0(0)	0.00000	0.00000	0	14580	0	293601392+	287677558+	0	0	120531966+	0	3687651+	0.0	
819+	ClientMach X	12.0	02:11.00	12	6	243+	78M+	0B	42M-	819	1	sleeping	*0(1)	0.13905	0.00000	501	398990+	451	410035+	34794+	4875633+	811816+	825853+	435	4337+	12.1	
427+	ysmond	10.2	00:38.33	3	2	22	1937K	0B	816K	427	1	sleeping	0(1156)	0.00000	0.24576	0	13297+	122	998528+	512258+	2485786+	858408+	49518+	8	12	10.2	
38720+	top	8.6	00:09.29	2/1	0	47+	17M	0B	2272K	38720	38318	running	*0(1)	0.00000	0.00000	0	64774+	144	2777915+	1388441+	310868+	1883663+	8777+	10	2	8.6	
865+	Boom 3D	8.5	19:23.72	9	1	374	69M	0B	62M	866	1	sleeping	*0(378)	0.00000	0.00000	501	63588	617	19342	42012	1440760+	1487759+	11144	28	8.5		
198+	coreaudio	7.3	17:28.02	11	3	969	27M	0B	15M	198	1	sleeping	*0(1)	0.00000	0.00079	202	36443	405	183640	168407+	5697850+	1892951+	3006330+	1843	296109+	8.2	
10000+	Code Helper	5.7	13:46.00	20	1	82	826M+	0B	360M	7844	7844	sleeping	*0(1)	0.00000	0.00000	501	416582	526	2776707+	550651+	14238916+	10850733+	7139057+	38	127884+	6.3	
35169+	Activity Mon	5.6	00:15.79	8	5	508	48M	96K	29M	35169	1	sleeping	0(08)	0.24730	0.00000	501	14145+	315	286387+	119507+	131780+	352111+	58000+	80	381+	5.6	
820+	ClientMach X	5.2	00:53.64	12	5	301+	70M+	0B	50M	820	1	sleeping	*0(1)	0.00039	0.12217	501	100000+	675	82114+	12516+	829211+	194021+	322514+	1307	1453	5.2	
22226+	AdobeAcrobat	3.2	02:38.55	15	4	362	256M	0B	183M	22226	1	sleeping	0(44)	0.04619	0.00000	501	187223	2670	2120189+	1045744+	234976+	6384473+	816488+	11139	86438+	4.1	
11525+	Vivaldi	2.7	09:19.94	43	1	1056	492M+	0B	240M	11525	1	sleeping	*0(1577+3)	0.00000	0.00000	501	3973291+	15503+	14617625+	8701258+	18138195+	46685656+	11381408+	18507	9134	2.7	
837+	iStat Menus	2.5	05:04.66	5	3	332	47M	3088K	23M	837	1	sleeping	*0(1)	0.04481	0.00000	501	1108400+	478	2021779+	493473+	1124482+	4591505+	1775031+	372	8879	2.5	
319+	iStatMenusDn	2.2	02:27.74	5	4	67	20M	0B	2704K	319	1	sleeping	*0(1)	0.00000	0.00016	0	14075	175	218553	182626+	18466360+	159000+	265432+	128	3474	2.2	
37637+	Terminal	1.9	00:13.29	11	4	390-	181M+	37M	9228K	37637	1	sleeping	*0(47)	0.00122	0.00000	501	107792+	105	187792+	38796+	241459+	352109+	58591+	1823	145+	1.9	
457+	distnoted	1.3	01:12.33	4	3	529	3985K	0B	1216K	457	1	sleeping	*0(1)	0.00000	0.03035	501	10945	125	31740+	83304+	1304105+	111227+	268239+	2	0	1.3	
10412+	Language ser	1.2	03:42.86	16	0	187	219M	0B	77M	7844	10303	sleeping	*0(1)	0.00000	0.00000	501	5395582+	734811	27031	13467	10087105+	13799+	6612197+	93	11684	1.2	
11554+	Vivaldi Help	1.0	01:43.49	19	1	134	83M+	0B	22M	11525	11525	sleeping	*0(444634+)	0.00000	0.00000	501	231725+	1236	5133025+	4260193+	7055382+	14263238+	4216699+	20	574	1.0	
944+	Bartender 5	0.9	01:20.08	7	4	295	34M+	1520K-	17M	944	1	sleeping	*0(250)	0.00433	0.00000	501	396541+	277809+	687524+	132828+	453529+	1640850+	533655+	295	3119	0.9	
159+	launchservic	0.7	00:33.66	4	3	603	6465K	0B	928K	159	1	sleeping	*0(589467+)	0.00000	0.01114	0	14574	146	996826+	828849+	2510904+	1608622+	754389+	45	85	0.7	
22001+	Vivaldi Help	0.7	00:35.01	19	1	272	370M+	0B	72M-	11525	11525	sleeping	*0(6403+)	0.00000	0.00000	501	285233+	1328	405961+	173892+	344414+	497283+	309177+	116	4836	0.7	
655+	AXVisualSupp	0.6	03:07.10	5	1	224	15M	0B	8000K	655	1	sleeping	*0(2032)	0.00000	0.00000	501	55234	276	3378691+	848914+	3415594+	7759373+	1707919+	199	28348+	0.7	
7844+	Electron	0.6	04:26.07	43	1	558	135M	0B	57M	7844	1	sleeping	*0(1489)	0.00000	0.00000	501	402614	1491	5609603+	1159796+	2990173+	11865215+	2465741+	1922	4299+	0.7	
170+	distnoted	0.6	00:40.57	3	2	198	2593K	0B	1184K	170	1	sleeping	*0(1)	0.00000	0.01404	241	3722	124	728305+	114087+	913500+	845455+	359108+	10	0	0.6	
817+	mongod	0.6	01:32.48	34	1	82	41M	0B	29M-	817	1	sleeping	*0(1)	0.00000	0.00000	501	33805+	264	99108+	49573+	925605+	85637+	628825+	2584	56554+	0.8	
863+	AlDente	0.4	00:41.91	4	2	211	24M+	0B-	14M	863	1	sleeping	*0(1956)	0.00478	0.00000	501	35477+	516	250835+	151470+	359360+	1059165+	250625+	351	4935	0.4	
7855+	Code Helper	0.4	11:09.88	20	1	332	324M	0B	65M	7844	7844	sleeping	*0(4)	0.00000	0.00000	501	1376139	623	2730045+	1177802+	4483541+	5891217+	3705227+	208	23742+	0.4	
13064+	Vivaldi Help	0.4	12:11.07	20	1	194	879M+	0B	677M-	11525	11525	sleeping	*0(560464+)	0.00000	0.00000	501	1083811+	1535	4475124+	1638300+	513661+	8883171+	4563265+	228	42004	0.4	
865+	iTranslate	0.3	00:10.51	6	2	253+	36M+	0B	28M	865	1	sleeping	*0(78)	0.00000	0.00000	501	13282+	419	3951	1338	188174+	98229+	32113+	294	2813	0.3	
655+	ControlCentr	0.3	01:38.68	7	2	628	38M	16K	21M	656	1	sleeping	*0(42445)	0.00109	0.00000	501	21946	694	769426+	160975+	751918+	2013060+	474998+	2373	8755	0.3	
507+	RTProtection	0.3	01:40.37	18	4	405	111M	0B	90M	507	1	sleeping	*0(1)	0.00000	0.00000	0	177276+	536	4632057+	2323690+	1077925+	2503311+	610084+	1426	30534	0.3	
1122+	fseventsd	0.2	01:02.55	18	1	233	4321K	0B	1200K	122	1	sleeping	*0(1)	0.00000	0.00000	0	173218+	128	310358+	116561+	3017856+	544434+	407227+	37	8726+	0.3	
21675+	WhatsApp	0.2	02:01.15	15	3	1309	220M	0B	191M	21675	1	sleeping	*0(3)	0.00045	0.00000	501	141142	2062	3086249+	313227+	1699710+	5801400+	649946+	1075	4074	0.3	
14	launchd	0.2	02:20.01	2	1	3908	20M	0B	5216K	1	0	sleeping	0(0)	0.00000	0.00411	0	33800	53	383242+	383624+	2408551+	1302953+	534001+	1079	150	0.2	
35131+	Vivaldi Help	0.2	00:20.74	20	1	252	293M	0B	132M	11525	11525	sleeping	*0(10013+)	0.00000	0.00000	501	18205	1326	98020+	48263+	184947+	220777+	159697+	12	1346+	0.3	
337+	com.apple.Dr	0.2	01:50.72	7	5	1460	38M	0B	5200K	337	1	sleeping	0(1)	0.00000	0.00079	270	95215+	82	370763+	353588+	2646278+	1135584+	1790530+	759	111792+	0.2	
226+	symptomsd	0.1	00:41.45	6	5	142	7137K	0B	2570K	226	1	sleeping	0(3993)	0.00000	0.00000	24	10341	207	35631	20941	2081312+	109058	770143+	357	252	0.1	
10202+	TabNine	0.1	00:25.51	65	1	175	37M	0B	26M	7844	10173	sleeping	*0(1)	0.00000	0.00000	501	9922	173	16643	8206	1807070+	290682+	710823+	160	79977+	1.1	
7846+	Dropbox	0.1	03:42.07	147	2	775	620M	0B	460M	7846	1	sleeping	*0(1723)	0.00000	0.00000	501	338688	23306	167004+	51487	5567371+	2614027+	2632813+	1915	17696+	0.3	
872+	com.applehou	0.1	00:28.07	3	2	50	4577K	0B	1328K	872	1	sleeping	*0(1)	0.00000	0.00360	0	215026	147	986220+	360342+	750955+	1209001+	460899+	1	314	0.1	
115+	logd	0.1	00:40.82	4	3	2278	22M	0B	25M	118	1	sleeping	*0(1)	0.00000	0.00042	0	41867	126	201309+	326590+	1844004+	421389+	649481+	67	24	0.1	

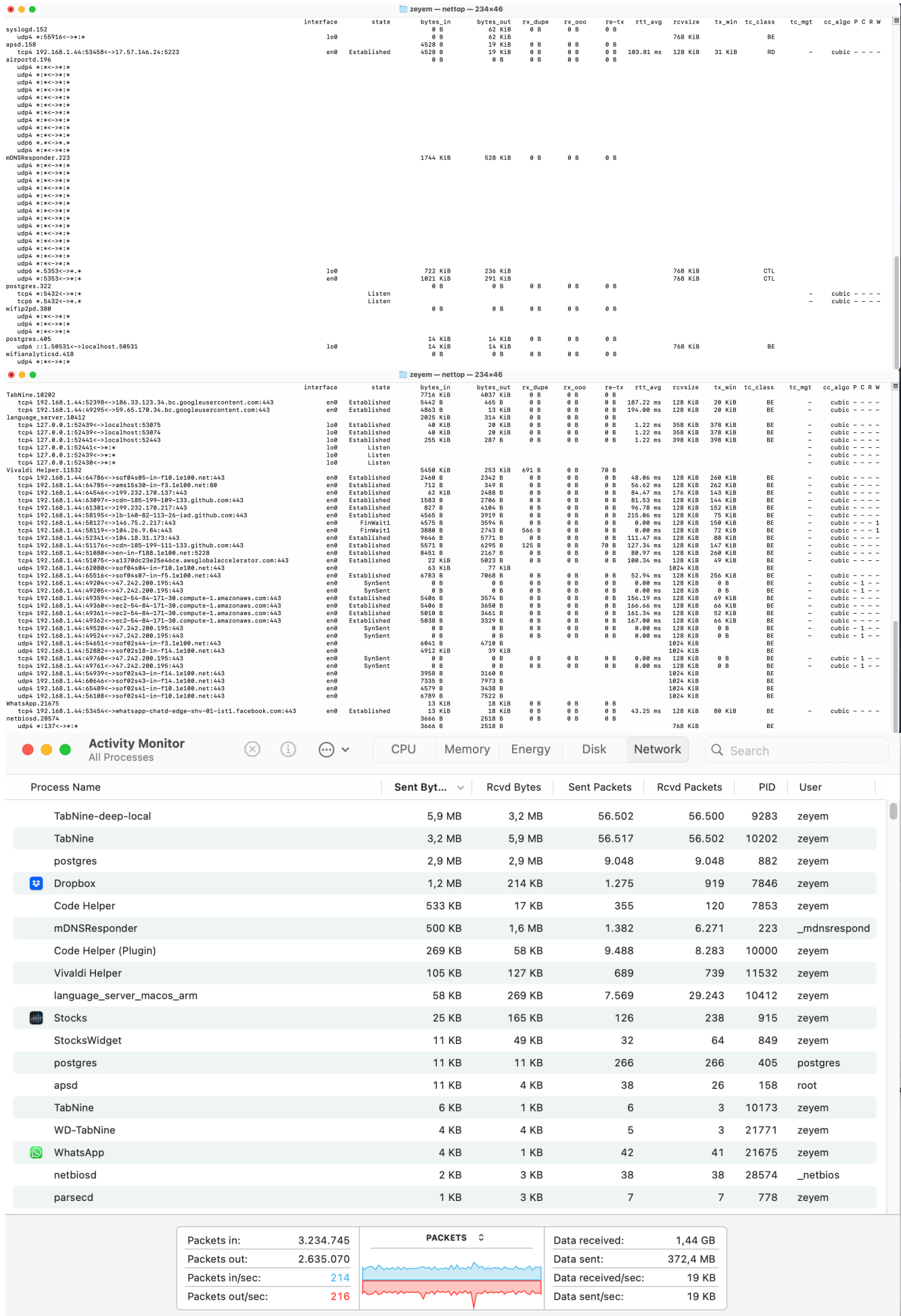
1.3. Monitoring Network Connections

Tracking network activity is essential for identifying unauthorized connections and monitoring resource usage. Below are alternatives to **netstat** for macOS.

1.3.1 nettop (Real-time Network Monitoring)

Run in Terminal: `nettop`

- Displays active network connections in real-time.



1.3.2 lsof (List Open Network Connections)

Run in Terminal: `sudo lsof -i -P`

Lists processes utilizing network connections, including ports and protocols.

```

zezem@zeys-MacBook-Pro-1894 ~ % nettop
zezem@zeys-MacBook-Pro-1894 ~ % sudo lsof -i -P
Password:
COMMAND      PID      USER   FD   TYPE    DEVICE  SIZE/OFF  NODE NAME
launchd        1       root   30u  IPv4  0x4c4df212d64222cb    0t0    UDP *:138
launchd        1       root   31u  IPv4  0x5b8c3a509a6f7b06    0t0    UDP *:137
configd     129       root   25u  IPv6  0x648f4dc3d20816a    0t0  ICMPV6 **:
syslogd     152       root    9u  IPv4  0xdf195c1d90e1c42c    0t0    UDP *:55916
airportd     196       root    3u  IPv4  0x325ce45c2e11d913    0t0    UDP **:
airportd     196       root    9u  IPv6  0x43a16a19d47994a7    0t0    UDP **:
airportd     196       root   11u  IPv4  0x336d29093664ffaa    0t0    UDP **:
airportd     196       root   12u  IPv4  0x6ce0d26ede42d368    0t0    UDP **:
airportd     196       root   13u  IPv4  0x182485173d6bc600    0t0    UDP **:
airportd     196       root   14u  IPv4  0xb4e4a2e31a10c94c    0t0    UDP **:
airportd     196       root   16u  IPv4  0xe815edc57b3e222    0t0    UDP **:
airportd     196       root   19u  IPv4  0xac46559a3913fdc8    0t0    UDP **:
airportd     196       root   20u  IPv4  0x1db648389da5a26f    0t0    UDP **:
airportd     196       root   22u  IPv4  0xc7591131d872ace    0t0    UDP **:
airportd     196       root   23u  IPv4  0xb022235ce58d977c    0t0    UDP **:
airportd     196       root   25u  IPv4  0xd73d911ad4df81a    0t0    UDP **:
mDNSRespo    223  _mdnsresponder    6u  IPv4  0x6c1343e783dcb283    0t0    UDP *:5353
mDNSRespo    223  _mdnsresponder    7u  IPv6  0x86efa3249a3949ba    0t0    UDP *:5353
mDNSRespo    223  _mdnsresponder   14u  IPv4  0xb9d41127b0832591    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   20u  IPv4  0xfe94e8015622a011    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   25u  IPv4  0x7e08d77cfb2d351c    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   29u  IPv4  0x4803e6e67c994ecb    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   39u  IPv4  0x18035bb76c8bd00c    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   42u  IPv4  0x73ef4761c51fe789    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   44u  IPv4  0x4b041b515c390dd    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   48u  IPv4  0x5d97183803dbecc1    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   51u  IPv4  0x678c85ee42fab702    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   54u  IPv4  0x5075db9548a7d327    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   64u  IPv4  0xebd21abc7778a0d6    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   68u  IPv4  0x33e5a3d14c5ec650    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   70u  IPv4  0x55c3eac71a193c90    0t0    UDP **:
mDNSRespo    223  _mdnsresponder   71u  IPv4  0xacfd1d7bba965e7    0t0    UDP **:
postgres    322      postgres    7u  IPv6  0xbf68b27a935fa3a5    0t0    TCP *:5432 (LISTEN)
postgres    322      postgres    8u  IPv4  0x46ca8c037c296a6c    0t0    TCP *:5432 (LISTEN)
postgres    322      postgres   10u  IPv6  0xdb6fcc4783e71fa5    0t0    UDP localhost:50531->localhost:50531
wifip2pd    380       root    4u  IPv4  0xd9e37a9ebbf4f039    0t0    UDP **:
wifip2pd    380       root    6u  IPv4  0xbe6519b66096ffd1    0t0    UDP **:
wifip2pd    380       root    9u  IPv4  0xfcdcf12e1a83a52a    0t0    UDP **:
postgres    401      postgres   10u  IPv6  0xdb6fcc4783e71fa5    0t0    UDP localhost:50531->localhost:50531
postgres    402      postgres   10u  IPv6  0xdb6fcc4783e71fa5    0t0    UDP localhost:50531->localhost:50531
postgres    403      postgres   10u  IPv6  0xdb6fcc4783e71fa5    0t0    UDP localhost:50531->localhost:50531
```

1.3.3 ifconfig (Check Network Interfaces)

Run in Terminal: `ifconfig`

```
zeyem@zeyem-MacBook-Pro-1894 ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM, TXCSUM, TXSTATUS, SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anp1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ce:ef:5b:92:bd:d4
    media: none
    status: inactive
anp0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ce:ef:5b:92:bd:d3
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ce:ef:5b:92:bd:b3
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ce:ef:5b:92:bd:b4
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=460<TS04, TS06, CHANNEL_IO>
    ether 36:cd:21:cf:32:40
    media: autoselect <full-duplex>
    status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=460<TS04, TS06, CHANNEL_IO>
    ether 36:cd:21:cf:32:44
    media: autoselect <full-duplex>
    status: inactive
ap1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6460<TS04, TS06, CHANNEL_IO, PARTIAL_CSUM, ZEROINVERT_CSUM>
    ether e2:08:e9:b0:df:73
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect (none)
    status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6460<TS04, TS06, CHANNEL_IO, PARTIAL_CSUM, ZEROINVERT_CSUM>
    ether 9e:01:4b:2e:00:7a
    inet6 fe80::10f7:cefb:da89:484a%en0 prefixlen 64 secured scopeid 0xb
    inet 192.168.1.44 netmask 0xfffff00 broadcast 192.168.1.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=6460<TS04, TS06, CHANNEL_IO, PARTIAL_CSUM, ZEROINVERT_CSUM>
    ether 9e:01:4b:2e:00:7a
    inet6 fe80::10f7:cefb:da89:484a%bridge0 prefixlen 64 secured scopeid 0xb
    inet 192.168.1.44 netmask 0xfffff00 broadcast 192.168.1.255
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect
    status: active
```

Displays details on active network interfaces, IP addresses, and data transmission.

		zeyem — -zsh — 162x57		~ — -zsh	
language_ 9454	zeyem	8u	IPv4 0x635ee86cd372fb32	0t0	TCP localhost:52172 (LISTEN)
language_ 9454	zeyem	15u	IPv4 0x6b577debbb2707a7	0t0	TCP localhost:52174 (LISTEN)
Code\x20H 10000	zeyem	34u	IPv4 0xa5b398644b2b21fc	0t0	TCP localhost:52443->localhost:52441 (ESTABLISHED)
Code\x20H 10000	zeyem	59u	IPv4 0x2e2246a5ebc6d2a3	0t0	TCP localhost:53074->localhost:52439 (ESTABLISHED)
Code\x20H 10000	zeyem	60u	IPv4 0x33ab45b78758bee0	0t0	TCP localhost:53075->localhost:52439 (ESTABLISHED)
TabNine 10173	zeyem	17u	IPv4 0x7fb7d30551b5717a	0t0	TCP 192.168.1.44:53671->59.65.170.34.bc.googleusercontent.com:443 (ESTABLISHED)
TabNine 10202	zeyem	15u	IPv4 0x490dcaa9dfbf4b2	0t0	TCP 192.168.1.44:52398->186.33.123.34.bc.googleusercontent.com:443 (ESTABLISHED)
TabNine 10202	zeyem	18u	IPv4 0xee921a3b29866e9d	0t0	TCP 192.168.1.44:49295->59.65.170.34.bc.googleusercontent.com:443 (ESTABLISHED)
TabNine 10202	zeyem	19u	IPv4 0xeccf7b4f727af2b1	0t0	TCP localhost:54719->localhost:49152 (ESTABLISHED)
TabNine 10202	zeyem	24u	IPv4 0xeccf7b4f727af2b1	0t0	TCP localhost:54719->localhost:49152 (ESTABLISHED)
TabNine 10202	zeyem	26u	IPv4 0x199e92849260db3	0t0	TCP localhost:1123 (LISTEN)
TabNine 10202	zeyem	61u	IPv4 0x8c10fffd1db026e1	0t0	TCP localhost:52411->localhost:49152 (ESTABLISHED)
TabNine 10202	zeyem	62u	IPv4 0x8c10fffd1db026e1	0t0	TCP localhost:52411->localhost:49152 (ESTABLISHED)
language_ 10412	zeyem	4u	IPv4 0x875308d01e3909ea	0t0	TCP localhost:52430 (LISTEN)
language_ 10412	zeyem	8u	IPv4 0x1d7280c977eecccb	0t0	TCP localhost:52439 (LISTEN)
language_ 10412	zeyem	15u	IPv4 0xe36d2acd2eaadb1	0t0	TCP localhost:52441 (LISTEN)
language_ 10412	zeyem	20u	IPv4 0x6885671ae735143e	0t0	TCP localhost:52439->localhost:53074 (ESTABLISHED)
language_ 10412	zeyem	21u	IPv4 0x1935cc6b2ce460b6	0t0	TCP localhost:52441->localhost:52443 (ESTABLISHED)
language_ 10412	zeyem	26u	IPv4 0xba2e888e5dfd2a92	0t0	TCP localhost:52439->localhost:53075 (ESTABLISHED)
Vivaldi 11532	zeyem	19u	IPv4 0x85011bbf50b13cec	0t0	TCP 192.168.1.44:50480->47.242.200.195:443 (SYN_SENT)
Vivaldi 11532	zeyem	22u	IPv4 0x18e0eee2f1f068f7	0t0	TCP 192.168.1.44:51080->en-in-f188.1e100.net:5228 (ESTABLISHED)
Vivaldi 11532	zeyem	26u	IPv4 0x8b84a8a793fec535	0t0	TCP 192.168.1.44:50477->47.242.200.195:443 (SYN_SENT)
Vivaldi 11532	zeyem	28u	IPv4 0xf869e3e7c8ecf2c3	0t0	TCP 192.168.1.44:50703->47.242.200.195:443 (SYN_SENT)
Vivaldi 11532	zeyem	29u	IPv4 0x41a16f924cd94fc2	0t0	TCP 192.168.1.44:51075->a1370dc23e25e46ce.amazonaws.com:443 (ESTABLISHED)
Vivaldi 11532	zeyem	31u	IPv4 0x8c8e90158379e1ea	0t0	TCP 192.168.1.44:50164->47.242.200.195:443 (SYN_SENT)
Vivaldi 11532	zeyem	32u	IPv4 0xdee621a5cb50faca	0t0	TCP 192.168.1.44:50702->47.242.200.195:443 (SYN_SENT)
Vivaldi 11532	zeyem	33u	IPv4 0x587fc711d4967ca8	0t0	TCP 192.168.1.44:50165->47.242.200.195:443 (SYN_SENT)
Vivaldi 11532	zeyem	34u	IPv4 0x6b30c02a150d1974	0t0	TCP 192.168.1.44:51176->cdn-185-199-111-133.github.com:443 (ESTABLISHED)
Vivaldi 11532	zeyem	35u	IPv4 0x7ceb9f3ef0d8cf48	0t0	TCP 192.168.1.44:64546->199.232.170.137:443 (ESTABLISHED)
Vivaldi 11532	zeyem	36u	IPv4 0xb454e96cf4f3fc92	0t0	UDP 192.168.1.44:63455->sof02s45-in-f10.1e100.net:443
Vivaldi 11532	zeyem	37u	IPv4 0x249580eeaa664af	0t0	TCP 192.168.1.44:58195->1b-140-82-113-26-1ad.github.com:443 (ESTABLISHED)
Vivaldi 11532	zeyem	39u	IPv4 0x4b5a3479c8a9764f	0t0	TCP 192.168.1.44:64785->ams15s30-in-f3.1e100.net:80 (ESTABLISHED)
Vivaldi 11532	zeyem	40u	IPv4 0x90230cac09d7af2a	0t0	TCP 192.168.1.44:52341->104.18.31.173:443 (ESTABLISHED)

```

Vivaldi 11532 zeyem 70u IPv4 0x2b578410b8f86571 0t0 TCP 192.168.1.44:50753->13.107.246.45:443 (ESTABLISHED)
Vivaldi 11532 zeyem 44u IPv4 0xe03c3c451ac2782d 0t0 TCP 192.168.1.44:64786->sof04s05-in-f10.1e100.net:443 (ESTABLISHED)
Vivaldi 11532 zeyem 45u IPv4 0x3c66ccf77be9bcba 0t0 UDP 192.168.1.44:64114->sof02s43-in-f14.1e100.net:443
Vivaldi 11532 zeyem 47u IPv4 0xc8a5bb915e8a7592 0t0 TCP 192.168.1.44:65516->sof04s07-in-f5.1e100.net:443 (ESTABLISHED)
Vivaldi 11532 zeyem 52u IPv4 0xd880e5be3294e3ca 0t0 TCP 192.168.1.44:61301->199.232.170.217:443 (ESTABLISHED)
Vivaldi 11532 zeyem 53u IPv4 0x5821af602ec181ee 0t0 TCP 192.168.1.44:51595->ams15s30-in-f3.1e100.net:80 (CLOSED)
Vivaldi 11532 zeyem 54u IPv4 0x57fab1203e92ff14 0t0 UDP 192.168.1.44:56108->sof02s41-in-f10.1e100.net:443
Vivaldi 11532 zeyem 56u IPv4 0x9b2f9a53548f505b 0t0 TCP 192.168.1.44:63097->cdn-185-199-109-133.github.com:443 (ESTABLISHED)
Vivaldi 11532 zeyem 60u IPv4 0x4d3afc0c1ca6709a 0t0 TCP 192.168.1.44:51596->sof02s45-in-f14.1e100.net:443 (CLOSED)
netbiosd 28574 _netbios 3u IPv4 0x5b8c3a509a6f7b06 0t0 UDP *:137
netbiosd 28574 _netbios 4u IPv4 0x4c4df212d64222cb 0t0 UDP *:138
WD-TabNin 44970 zeyem 15u IPv4 0x77bae7d683bee1e5 0t0 TCP 192.168.1.44:57361->59.65.170.34.bc.googleusercontent.com:443 (ESTABLISHED)
zeyem@zeys-MacBook-Pro-1894 ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM, TXCSUM, TXSTATUS, SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xff000000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anp1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_IO>
    ether ce:ef:5b:92:bd:d4

```

2. Checking and expanding your SWAP memory

macOS dynamically manages swap memory, but users can monitor and adjust its behavior. Swap memory (virtual memory) is a portion of disk storage that macOS uses as an **extension of RAM** when physical memory is full. When the system runs out of RAM, inactive memory pages are moved to the swap space to **free up RAM for active processes**, preventing system slowdowns and application crashes.

2.1 Checking Current Swap Usage

To check swap memory usage, run:

```

sysctl vm.swapusage
[zeyem@zeys-MacBook-Pro-1894 ~ % sysctl vm.swapusage
vm.swapusage: total = 1024.00M  used = 141.62M  free = 882.38M  (encrypted)

```

2.2 Checking System Memory Usage

Run: `vm_stat`

- This command provides details on **free memory, inactive memory, and swap usage**.

```

[zeyem@zeys-MacBook-Pro-1894 ~ % vm_stat
Mach Virtual Memory Statistics: (page size of 16384 bytes)
Pages free:                                9255.
Pages active:                             283170.
Pages inactive:                           283922.
Pages speculative:                         430.
Pages throttled:                           0.
Pages wired down:                         120119.
Pages purgeable:                           4422.
"Translation faults":                     116021366.
Pages copy-on-write:                       9876686.
Pages zero filled:                         48032960.
Pages reactivated:                         6673343.
Pages purged:                              793237.
File-backed pages:                        179740.
Anonymous pages:                          387782.
Pages stored in compressor:                710870.
Pages occupied by compressor:              315904.
Decompressions:                           3934932.
Compressions:                             6255980.
Pageins:                                  2980009.
Pageouts:                                 33458.
Swapins:                                  5546.
Swapouts:                                 23400.

```

2.3 Adjusting Swap Memory

macOS dynamically manages swap based on system requirements, but users can **enable or disable** it manually but it isn't recommended for most users. On macOS, swap memory works differently than on Windows or Linux. Virtual memory in macOS involves the use of:

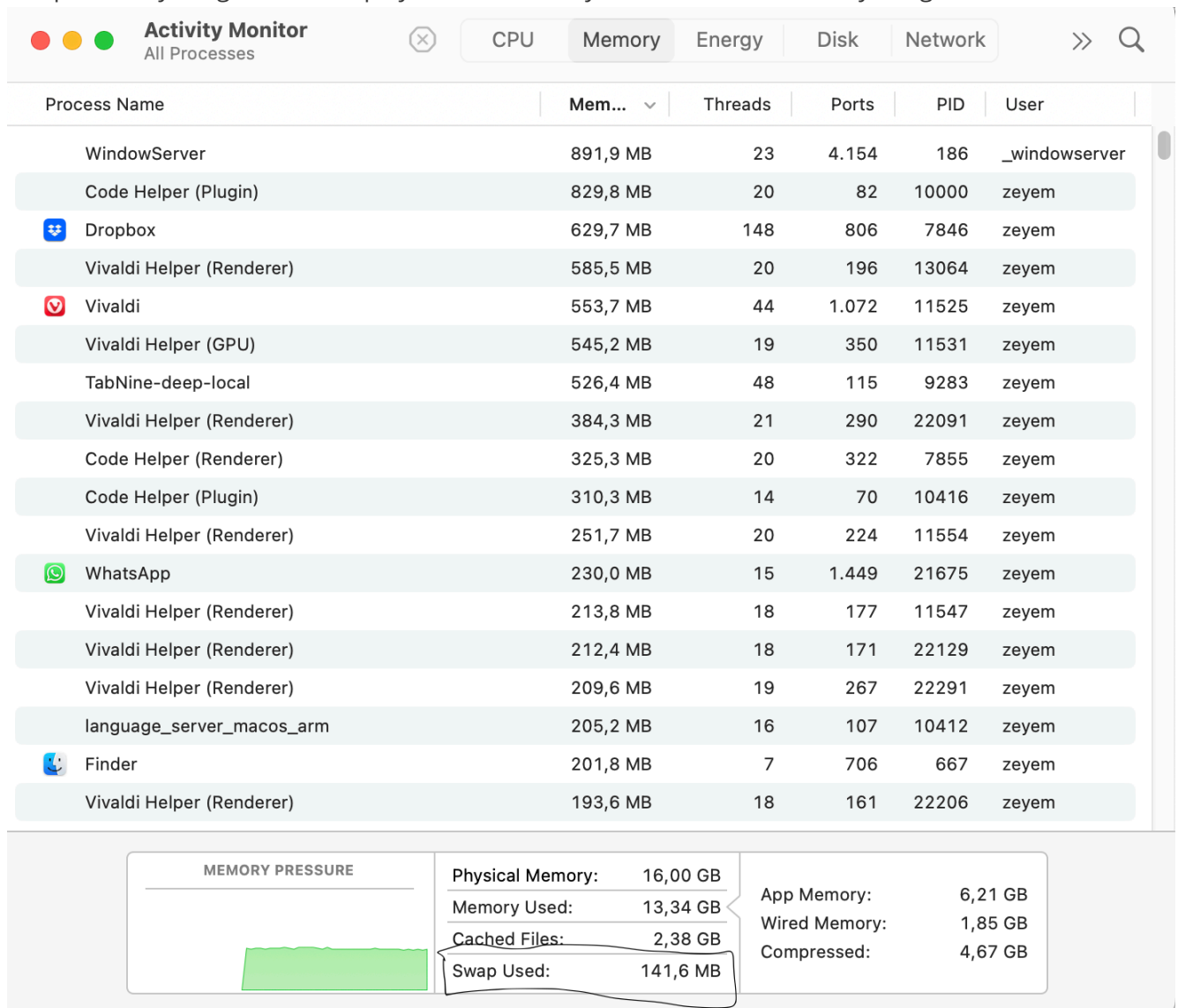
- **Memory Compression:** Compresses inactive memory pages' contents, reducing the need for swapping to disk and improving overall system performance.
- **Swap Space:** Serves as a last resort when other memory management techniques cannot handle memory demands.

Disable/Enable Swap (Not Recommended)

```
sudo launchctl unload -w /System/Library/LaunchDaemons/com.apple.dynamic_pager.plist
```

- macOS does not allow disabling swap this way in newer versions. It gives `Boot-out failed: 5: Input/output error`
- In this case, we use `sudo purge` command to free up inactive memory and close unnecessary applications.

- Swap memory usage can be displayed in the Activity Monitor under Memory Usage:



The Swap Used value shows how much swap memory is currently being used.

Conclusion

This report provides an overview of how macOS handles **process monitoring, memory management, and network connections**. By using built-in and third-party tools like **Activity Monitor**, **top**, **nettop**, and **sysctl**, users can efficiently manage system resources. Proper monitoring helps in **optimizing performance and ensuring system stability**.

References

- Apple Support Documentation: <https://support.apple.com>
- macOS Terminal Commands: <https://ss64.com/osx/>